DecisionRL: Enhancing Decision Calibration through Reinforcement Learning

Reinforcement Learning Course Project Proposal

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Research Question

As large language models evolve from question answerers to decision makers, efficient and reliable decision-making becomes increasingly important. While System 2-style models emphasize deliberate, multi-step reasoning, System 1-style models aim to provide fast, intuitive judgments. Jev introduces a complementary paradigm of structured probabilistic decisions, producing explicit probability distributions over candidate actions instead of lengthy reasoning trajectories. Inspired by this paradigm, our project explores how **Reinforcement Learning for Calibrated Decisions (RLCD)** can enhance decision accuracy, probability calibration, and reliability through a reproducible training framework.

Related Work

Jev introduces System 1-style probabilistic decision-making and RLCD, while **Microsoft-Decision-1** demonstrates the potential of efficient decision models. Within the open-source ecosystem, **Kev** and **Laya** illustrate different training approaches for typed-decision models: Kev primarily combines supervised learning with post-hoc calibration, whereas Laya incorporates RLCD to improve probability calibration. These approaches motivate our investigation of what RLCD adds to supervised training and how its optimization can be improved. Our project will develop a reproducible two-stage training framework to investigate the benefits of RLCD beyond supervised fine-tuning and evaluate strategies for improving decision accuracy and probability calibration.

Planned Approach

We plan to develop **DecisionRL**, a decision model based on **EuroBERT**, through supervised fine-tuning (SFT) followed by RLCD. We will build our own training and evaluation pipeline to compare an RLCD baseline with proposed optimization variants. The work will comprise two components:

- **Data Construction and Curation.** We will collect and curate decision datasets with hard and soft labels, using a common processing pipeline for training and evaluation. Hard-labeled data will support supervised fine-tuning to establish foundational classification capabilities, while soft-labeled data will provide probabilistic supervision for RLCD. We will maintain separate training and test splits to support consistent comparisons across training stages and optimization variants.
- **Two-Stage Training and RLCD Optimization.** Starting from EuroBERT, we will perform supervised fine-tuning on hard-labeled decision data to obtain our own SFT checkpoint. We will then establish an RLCD baseline based on Laya's RLCD formulation, using soft-label supervision. Using the same SFT checkpoint, we will compare this baseline with improved RLCD optimization strategies for more accurate and better-calibrated decisions.

Evaluation Plan

We will evaluate DecisionRL on hard-label benchmarks (e.g. Banking77) and soft-label benchmarks (e.g. Typed Decisions), comparing it with representative decision models (e.g. Jev). The evaluation will use **Accuracy**, **Macro F1 Score**, **Expected Calibration Error**, and **Brier Score** to assess the foundational decision-making capabilities established through SFT, with particular emphasis on analyzing how RLCD and our proposed improvements affect probability calibration and decision-making behavior.

Required Resources and Division of Responsibilities

Resources: No additional equipment or specialized resources are required.

Preliminary Division of Responsibilities: Both team members will jointly participate in project discussions, research planning, and RLCD methodological design. Yanyun LIU will primarily contribute to data collection, synthesis, and quality control, while Taihua Chen will focus more on model and algorithm optimization.